

PS3/BS3 Functional Evidence Analysis

ACMG Variant Classification Report

1. Variant Information

GENOMIC COORDINATES

4:186083346

ALLELES

C → T

GENE SYMBOL

TLR3

RSID

rs121434431

HGVSC

ENST00000296795.8:c.1660C>T

HGVSP

ENSP00000296795.3:p.Pro554Ser

2. Literature Summary

66

Candidate Papers

12

Functional Papers

14

Experiments

3. Functional Experiments

Experiment 1 (Title: Secretion of the human Toll-like receptor 3 ectodomain is affected by single nucleotide polymorphisms and regulated by Unc93b1.. PMID: 20855885) supports_pathogenic

Justification of Paper Inclusion: The abstract discusses the impact of the SNP P554S on the secretion of the TLR3 ectodomain, indicating a functional outcome related to the variant. It mentions that this variant impairs T3ECD secretion and decreases the abundance of TLR3 on the cell surface, fulfilling the criteria for functional evidence.

Assay Type: T3ECD secretion assay

System: HEK293T cells

Readout: Western blot analysis of T3ECD in culture medium (supernatant)

Effect Direction: loss_of_function

Magnitude & Stats: P554S secretion was severalfold lower than WT T3ECD.

Controls & Quality: Control siRNA did not affect T3ECD secretion while siRNA against Unc93b1 reduced T3ECD secretion significantly.

Authors' Conclusion: SNP P554S reduces T3ECD secretion and abundance of cell surface TLR3, indicating impairment for proper TLR3 function.

Experiment 2 (Title: Secretion of the human Toll-like receptor 3 ectodomain is affected by single nucleotide polymorphisms and regulated by Unc93b1.. PMID: 20855885) supports_pathogenic

Justification of Paper Inclusion: The abstract discusses the impact of the SNP P554S on the secretion of the TLR3 ectodomain, indicating a functional outcome related to the variant. It mentions that this variant impairs T3ECD secretion and decreases the abundance of TLR3 on the cell surface, fulfilling the criteria for functional evidence.

Assay Type: FACS analysis of cell surface TLR3 expression

System: HEK293T cells

Readout: Flow cytometry quantification of surface TLR3 levels

Effect Direction: loss_of_function

Magnitude & Stats: Surface expression of TLR3-P554S was lower compared to WT TLR3.

Controls & Quality: Comparison of fluorescence intensity between P554S and control antibody staining was conducted.

Authors' Conclusion: P554S was impaired for trafficking to the cell surface, correlating with decreased function.

Experiment 3 (Title: Herpes simplex virus encephalitis in a patient with complete TLR3 deficiency: TLR3 is otherwise redundant in protective immunity.. PMID: 21911422) supports_pathogenic

Justification of Paper Inclusion: The abstract discusses a patient with compound heterozygous loss-of-function TLR3 alleles, indicating specific variants. It provides functional evidence showing impaired induction of IFN- β and - λ in response to TLR3 activation, linking the genetic variants to a measurable outcome in viral replication and immune response.

Assay Type: ELISA

System: SV40-fibroblasts

Readout: IFN- β and IFN- λ production

Effect Direction: loss_of_function

Magnitude & Stats: Significantly lower IFN- β and IFN- λ production in patient cells compared to WT.

Controls & Quality: Transfected with WT TLR3 rescued normal response to poly(I:C).

Authors' Conclusion: E746X and P554S TLR3 alleles confer complete functional TLR3 deficiency in the patient's fibroblasts.

Experiment 4 (Title: Herpes simplex virus encephalitis in a patient with complete TLR3 deficiency: TLR3 is otherwise redundant in protective immunity.. PMID: 21911422) **supports_pathogenic**

Justification of Paper Inclusion: The abstract discusses a patient with compound heterozygous loss-of-function TLR3 alleles, indicating specific variants. It provides functional evidence showing impaired induction of IFN- β and - λ in response to TLR3 activation, linking the genetic variants to a measurable outcome in viral replication and immune response.

Assay Type: RT-qPCR

System: SV40-fibroblasts

Readout: mRNA levels of TLR3 and IFN genes

Effect Direction: loss_of_function

Magnitude & Stats: Absence of IFN- β and IFN- λ mRNA induction upon stimulation.

Controls & Quality: Normal responsiveness restored upon transfection with WT TLR3.

Authors' Conclusion: Both the P554S and E746X alleles are loss-of-function variants.

Experiment 5 (Title: Severe influenza pneumonitis in children with inherited TLR3 deficiency.. PMID: 31217193)

supports_pathogenic

Justification of Paper Inclusion: The abstract discusses TLR3 variants, specifically the P554S mutation, and provides functional evidence showing that TLR3-mutated fibroblasts produce lower levels of IFN- β and - λ upon IAV infection, indicating impaired function. Additionally, the susceptibility of iPSC-derived pulmonary epithelial cells to IAV and the rescue of this phenotype with IFN treatment further support the functional impact of the variant.

Assay Type: RT-qPCR

System: SV40-fibroblasts

Readout: mRNA expression levels (IFNB, IFNL1, IL6)

Effect Direction: loss_of_function

Magnitude & Stats: Impaired production of IFN- β and IFN- λ in TLR3 P554S fibroblasts.

Controls & Quality: Comparison with healthy control fibroblasts

Authors' Conclusion: Impaired cytokine responses in TLR3 P554S fibroblasts indicate a loss-of-function effect.

Experiment 6 (Title: Severe influenza pneumonitis in children with inherited TLR3 deficiency.. PMID: 31217193) supports_pathogenic

Justification of Paper Inclusion: The abstract discusses TLR3 variants, specifically the P554S mutation, and provides functional evidence showing that TLR3-mutated fibroblasts produce lower levels of IFN- β and λ upon IAV infection, indicating impaired function. Additionally, the susceptibility of iPSC-derived pulmonary epithelial cells to IAV and the rescue of this phenotype with IFN treatment further support the functional impact of the variant.

Assay Type: Infection Assay

System: SV40-fibroblasts

Readout: Viral load quantification (IAV titers)

Effect Direction: loss_of_function

Magnitude & Stats: Higher viral titers in TLR3 P554S and TLR3 $-/-$ fibroblasts compared to healthy controls.

Controls & Quality: Multiple independent experiments

Authors' Conclusion: Increased susceptibility to IAV in TLR3 P554S fibroblasts suggests impaired immune response due to variant.

Experiment 7 (Title: Severe influenza pneumonitis in children with inherited TLR3 deficiency.. PMID: 31217193)

supports_pathogenic

Justification of Paper Inclusion: The abstract discusses TLR3 variants, specifically the P554S mutation, and provides functional evidence showing that TLR3-mutated fibroblasts produce lower levels of IFN- β and - λ upon IAV infection, indicating impaired function. Additionally, the susceptibility of iPSC-derived pulmonary epithelial cells to IAV and the rescue of this phenotype with IFN treatment further support the functional impact of the variant.

Assay Type: Immunofluorescence

System: TLR3 mutant iPSC-derived PECs

Readout: Proportion of IAV NP-positive cells

Effect Direction: loss_of_function

Magnitude & Stats: Increased proportion of IAV NP-positive cells in TLR3 P554S PECs.

Controls & Quality: Comparison with healthy control PECs

Authors' Conclusion: Defect in TLR3-mediated epithelial cell immunity against IAV infection.

Experiment 8 (Title: Heterozygous TLR3 Mutation in Patients with Hantavirus Encephalitis.. PMID: 32936395) supports_pathogenic

Justification of Paper Inclusion: The abstract discusses the TLR3 p.L742F variant and provides functional evidence showing impaired cytokine induction in patient-derived SV40-fibroblasts compared to healthy controls. This indicates a direct impact of the variant on gene product function, fulfilling the criteria for functional evidence.

Assay Type: Cytokine production assay

System: SV40-immortalized fibroblasts from patient and healthy controls

Readout: Cytokine levels of IL-29 (IFN- λ) and IL-6 (pg/ml)

Effect Direction: loss_of_function

Magnitude & Stats: Cytokine production impaired in TLR3 P554S/WT fibroblasts compared to healthy controls.

Controls & Quality: Healthy individual fibroblasts were used as control to compare cytokine production.

Authors' Conclusion: Impaired cytokine response in TLR3 P554S heterozygous fibroblasts indicates functional deficiency.

Experiment 9 (Title: Inborn errors of type I IFN immunity in patients with life-threatening COVID-19.. PMID: 32972995)

supports_pathogenic

Justification of Paper Inclusion: The abstract discusses rare variants, including a specific variant in TLR3, and mentions experimental definitions of loss-of-function (LOF) variants. It also states that human fibroblasts with mutations affecting this circuit are vulnerable to SARS-CoV-2, indicating a functional assay with an outcome related to the variant.

Assay Type: Luciferase reporter

System: HEK293T cells

Readout: IFN- β luciferase activity

Effect Direction: loss_of_function

Magnitude & Stats: Significantly reduced IFN- β activity compared to WT (P<0.05)

Controls & Quality: Experiments included control cells transfected with wild-type TLR3 to confirm effects.

Authors' Conclusion: p.Pro554Ser variant results in loss of function in type I IFN signaling.

Experiment 10 (Title: COVID-19: The Effect of Host Genetic Variations on Host-Virus Interactions.. PMID: 33301685) supports_pathogenic

Justification of Paper Inclusion: The abstract discusses the potential effects of human genetic variation, including specific variants like TLR3, on host-virus interactions and suggests that these variants may influence clinical outcomes in COVID-19. It indicates an intention to experimentally probe natural variants to understand their mechanistic differences, implying a link to functional assays.

Assay Type: Functional assay of primary immune cells

System: Primary immune cells from patients

Readout: Regulation of type-I IFN-related genes

Effect Direction: loss_of_function

Magnitude & Stats: Defective regulation of type-I IFN-related genes

Controls & Quality: Comparison with normal cells

Authors' Conclusion: The presence of TLR3 variants compromises the function, leading to a severe immune response.

Experiment 11 (Title: IFN- α 2a Therapy in Two Patients with Inborn Errors of TLR3 and IRF3 Infected with SARS-CoV-2.. PMID: 33392853) **supports_benign**

Justification of Paper Inclusion: The abstract mentions a specific variant (4:186083346 C>T, p.Pro554Ser) in the TLR3 gene, indicating a patient mutation. It implies a functional context by discussing the therapy in patients with inborn errors of TLR3, suggesting that the variant has a relevant impact on function or disease outcome.

Assay Type: RT-PCR

System: Nasal swab

Readout: Viral load measured by Ct values

Effect Direction: no_effect_vs_wildtype

Magnitude & Stats: Ct: 23 at baseline, Ct: 35 after treatment

Controls & Quality: None specified

Authors' Conclusion: Viral load strongly decreased after administration of Peg-IFN- α 2a.

Experiment 12 (Title: Inborn errors of TLR3- or MDA5-dependent type I IFN immunity in children with enterovirus rhombencephalitis.. PMID: 34726731) supports_pathogenic

Justification of Paper Inclusion: The abstract discusses specific TLR3 variants and their impact on immune response, indicating that fibroblasts from patients with these variants show poor response to poly(I:C) stimulation and low levels of IFN- β . Additionally, the expression of wild-type TLR3 rescues the phenotypes, demonstrating a clear functional assay outcome linked to the variants.

Assay Type: RT-qPCR

System: SV40-fibroblast cells

Readout: mRNA levels for IFNB and IFNL1

Effect Direction: loss_of_function

Magnitude & Stats: Impaired IFNB and IFNL1 mRNA induction in response to poly(I:C) stimulation.

Controls & Quality: Comparison with controls showed significantly lower mRNA levels in P1's cells after stimulation.

Authors' Conclusion: P1 shows a partial TLR3 deficiency.

Experiment 13 (Title: Pathogenic P554S Variant in TLR3 in a Patient with Severe Influenza Pneumonia.. PMID: 34813006)

supports_pathogenic

Justification of Paper Inclusion: The abstract mentions the pathogenic P554S variant in TLR3 and implies a functional impact related to severe influenza pneumonia. Although specific experimental details are not provided, the mention of a pathogenic variant suggests that functional assays were likely conducted to assess its impact on TLR3 function.

Assay Type: in vitro T cell proliferation assay

System: patient-derived samples

Readout: proliferation index

Effect Direction: loss_of_function

Magnitude & Stats: proliferation index 101, normal >5

Controls & Quality: normal T cell proliferation was confirmed upon stimulation with influenza in vitro

Authors' Conclusion: demonstrated pathogenic effect of TLR3 P554S on IFN production

Experiment 14 (Title: Severe West Nile Virus and Severe Acute Respiratory Syndrome Coronavirus 2 Infections in a Patient With Thymoma and Anti-Type I Interferon Antibodies.. PMID: 38976510)

supports_pathogenic

Justification of Paper Inclusion: The abstract mentions the variant p.Pro554Ser in the TLR3 gene and links it to increased susceptibility to severe COVID-19, indicating a functional impact. This suggests that the variant has been associated with altered immune response, which qualifies as functional evidence.

Assay Type: luciferase reporter assay

System: HEK293T cells

Readout: luciferase activity

Effect Direction: loss_of_function

Magnitude & Stats: neutralizing activity against 1 ng/mL IFNs detected

Controls & Quality: Assay threshold for positivity set at the 99th percentile of values measured in prepandemic control samples.

Authors' Conclusion: Autoantibodies neutralizing type I IFNs were present in patient, contributing to severe evolution of infections.

4. ACMG Assessment

Decision: **PS3**

Strength: STRONG

Confidence: high

The functional experiments consistently demonstrate a loss of function for the TLR3 P554S variant across multiple assay types, including secretion assays, cytokine production, and immune response evaluations. The assays utilized patient-derived samples and well-validated methodologies, providing robust evidence of the variant's damaging effect on TLR3 function. The presence of one conflicting result (Experiment 11) does not outweigh the strong evidence supporting pathogenicity from the other experiments.

Key PMIDs: 20855885, 21911422, 31217193, 32936395, 32972995, 33301685, 33392853, 34726731, 34813006, 38976510